

Securing Your Place in the Review Network: A Dynamic Embeddedness-aware Graph Neural Network for Restaurant Survival Prediction

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Abstract

Restaurants, as small hospitality businesses, are inherently vulnerable, making accurate survival prediction crucial. Previous studies have demonstrated the significance of user reviews and incorporated diverse review-derived factors, yet they have largely overlooked the large-scale network formed by user–restaurant interactions. How restaurant survival is influenced by the review network remains insufficiently explored. To fill this gap, leveraging network embeddedness theory, we statistically analyze the impact of two dimensions of embeddedness, structural and positional, on each restaurant’s survival. Utilizing two real-world review datasets, the newly curated OpenRice and the well-established Yelp, our results reveal that a restaurant’s network embeddedness and its temporal evolution positively correlate with its survival. Building on this insight, we propose a Dynamic Embeddedness-aware Graph Neural Network, DyE-GNN, for restaurant survival prediction. DyE-GNN not only explicitly integrates network embeddedness theory to guide the model design but also leverages domain knowledge to enable robust adaptability. Extensive experiments on both datasets confirm the superiority of DyE-GNN, underscoring the importance of network embeddedness attention, temporal dynamics, and survival knowledge of peer restaurants. Visualizations further demonstrate that network embeddedness facilitates the identification of at-risk restaurants at the network margin. Our code and data are released at <https://github.com/zy199710/DyE-GNN>.

CCS Concepts

• **Information systems** → **Data mining; Social networks; • Applied computing** → *Sociology; Enterprise modeling.*

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Keywords

Restaurant survival prediction, Dynamic graph neural network, Network embeddedness, Online review network

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1 Introduction

Restaurants, as an important component of small hospitality businesses, represent a vital pillar of the country’s economy, yet remain inherently vulnerable. Unlike large-scale enterprises, this type of business possesses little capacity to withstand crises and is the first to bear the brunt of adverse impacts [15, 33]. Predicting restaurant survival and promptly providing warnings to operators appears to be an urgent necessity.

With the proliferation of online review platforms such as Yelp, Dianping, and OpenRice, user-posted reviews have proven valuable for predicting restaurant survival. Existing review-based survival prediction studies explored various types of review information, such as review rating [20, 24], review textual content [17, 18, 34], review photos [43], and user–restaurant interaction patterns [19, 22]. However, network embeddedness theory emphasizes that the behavior of individuals or organizations is influenced by the various relationships within the social network in which they are embedded. The survival of a restaurant can depend not only on its intrinsic qualities but also on its place in the broader review network, which remains largely underexplored in existing studies.

Therefore, in this paper, we introduce an innovative application of network embeddedness theory to quantify the impact of review networks on restaurant survival and to assess its implications for survival prediction models. We intend to solve two research questions: (1) *Does review network embeddedness affect restaurant survival?* (2) *How can review network embeddedness enhance restaurant survival prediction models?*

To address question 1, we statistically examine the relationship between network embeddedness and restaurant survival along the

two most representative dimensions: structural and positional [31]. Specifically, in annual review networks constructed from two real-world datasets (OpenRice and Yelp), we compute and analyze the distributional differences of these network embeddedness and their temporal evolution with restaurant survival, as well as their effects on the Kaplan–Meier (KM) survival curve. The details of our statistics are illustrated in the Motivation section. In conclusion, we have the following two observations: (1) Surviving and closed restaurants differ significantly in both dimensions of embeddedness and their temporal evolution. (2) Lower embeddedness and its temporal evolution result directly in lower survival curves. These findings suggest that network embeddedness is positively correlated with restaurant survival and motivate further exploration of its predictive utility.

Hence, as for question 2, we propose a Dynamic Embeddedness-aware Graph Neural Network model, DyE-GNN, for predicting restaurant survival in dynamic review networks. Unlike existing methods that treat restaurant and user nodes identically, DyE-GNN employs distinct aggregation strategies for each node type and integrates network embeddedness exclusively for restaurant nodes. To capture both short-term and long-term review dynamics and their effects on restaurant survival, DyE-GNN extends classical dynamic GNNs by encoding daily review fluctuations at the date-scale and annual network transformations at the year-scale. Furthermore, for the first time, DyE-GNN incorporates survival labels across multiple temporal spans, reflecting real-world prediction scenarios. Leveraging both a newly curated OpenRice dataset and the established Yelp benchmark, we construct dynamic review networks and perform extensive experiments. The results demonstrate that DyE-GNN outperforms all baseline methods across various evaluation metrics. Ablation and parameter analyses confirm that network embeddedness attention, temporal dynamics, and survival knowledge from restaurants with varying lifespans are all critical for accurate survival prediction. Finally, a model-mechanism visualization demonstrates how DyE-GNN effectively integrates network embeddedness and its temporal evolution to identify at-risk restaurants at the network’s margin more precisely.

Our literature review on restaurant survival prediction, and dynamic graph neural networks can be found in Appendix A.

The contributions of our paper are summarized as follows:

- We provide the first empirical validation that structural and positional embeddedness, as well as their temporal evolution, are positively correlated with restaurant survival, thereby contributing to restaurant survival analysis and prediction studies.
- We curate a new dataset from the OpenRice review website, in addition to the commonly used Yelp dataset, that supports restaurant survival prediction with graph-based methods.
- We propose a novel dynamic embeddedness-aware GNN model, DyE-GNN, to predict restaurant survival in real-world review scenarios. DyE-GNN introduces network embeddedness to guide the model construction and leverages domain knowledge to enable robust adaptability, contributing network-science inspired design to the GNN community.
- We provide practical restaurant management insights on how network embeddedness and review dynamics affect

survival, suggesting that managers should focus on securing the embeddedness of their restaurants in review networks, such as enhancing restaurant exposure, attracting highly active and influential reviewers, and maintaining a positive trend in review dynamics.

2 Preliminary

In this section, we introduce the necessary definitions and present our preliminary results to motivate the study.

2.1 Definitions

For each year timestamp, we define the network as follows:

Review Network. Let the review network be represented as a bipartite graph $\mathcal{G} = (\mathcal{V}_R \cup \mathcal{V}_U, \mathcal{E}, \mathcal{T}_E)$, while $v_r \in \mathcal{V}_R$ and $v_u \in \mathcal{V}_U$ denote restaurant and user nodes, respectively. An edge $(v_r, v_u) \in \mathcal{E}$ exists if user v_u has reviewed restaurant v_r . $\mathcal{T}_E \in \mathbb{R}^{|\mathcal{E}|}$ stores the corresponding calendar date for each review.

Dynamic Review Networks. Let $\mathcal{G}$ denote the static restaurant review network. The dynamic review networks are defined as a sequence of such networks over different timestamps: $\mathcal{G}^{dyn} = \{\mathcal{G}^T \mid T = T_1, T_2, \dots, T_n\}$. Formally, if a restaurant remains operational at both T_i and T_j for $i \neq j$, then $\mathcal{V}_R^{(T_i)} \cap \mathcal{V}_R^{(T_j)} \neq \emptyset$.

2.2 Motivation

Inspired by the success of Petryk et al. in analyzing multiple dimensions of network embeddedness within employee rating networks [31], we focus on the two key embeddedness dimensions: structural and positional, and compute them in the dynamic review networks. The specific calculations for each dimension are described as follows.

Structural Embeddedness. This dimension considers the extent to which an individual’s economic exchange behavior is situated within the context of ongoing interpersonal relationships [7]. In the review network, it captures two aspects: (1) the number of reviews a restaurant receives, and (2) the activity level of its reviewers, which together reflect the restaurant’s direct exposure to users [23]. A high structural embeddedness score indicates that a restaurant has received many reviews and is frequently reviewed by highly active users. Given a target restaurant v_r , the structural embeddedness is defined as the neighbor out-degree centrality:

$$SE(v_r) = \frac{\sum_{v_u \in N(v_r)} Deg_{v_u}}{|\mathcal{V}_R|}, \quad (1)$$

where Deg_{v_u} means the out-degree of the user node v_u and $N(v_r)$ denotes the neighbor set of v_r .

Positional Embeddedness. This metric quantifies the degree to which an entity is connected to other entities that are themselves highly embedded in the network [31]. For a review network, this embeddedness emphasizes a restaurant’s strategic position within the entire review network and its ability to disseminate information. Restaurants with high positional embeddedness resemble “hot” venues that attract influential reviewers. We calculate it by the

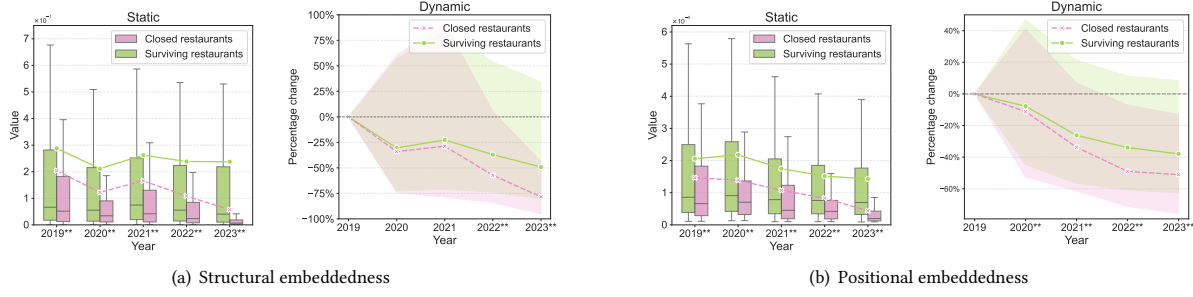


Figure 1: The distribution of network embeddedness for restaurants with different survival scenarios. We annotate the x-axis to indicate the statistical significance between the two annual distributions. In the box plot, the boundaries of each box correspond to the first and third quartiles, and a superimposed line traces the mean values. In the line plot, we calculate the percentage change relative to the first year as $(T_i - T_1)/T_1$, and the plotted line denotes the median percentage change for each year.

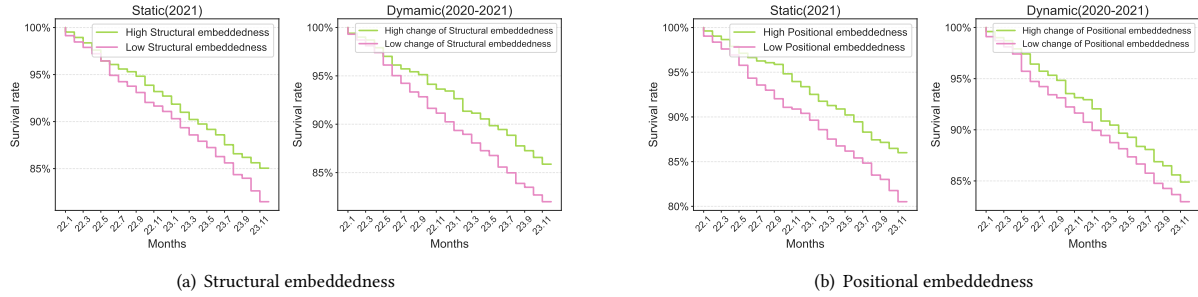


Figure 2: The effect of network embeddedness on restaurants' KM survival curves. For static network embeddedness, we divide the corresponding values of all restaurants into high-value and low-value groups and plot KM survival curves. For dynamic network embeddedness, we calculate the difference between the two years and divide them into high-difference and low-difference groups.

PageRank metric [25], which is defined as:

$$PE(v_r) = \frac{1 - \delta}{|\mathcal{V}_R \cup \mathcal{V}_U|} + \delta \sum_{v_u \in N(v_r)} \frac{PE(v_u)}{Deg(v_u)}, \quad (2)$$

where δ is the damping factor.

To examine the correlation between network embeddedness and restaurant survival, we conduct preliminary analyses on two datasets: OpenRice and Yelp. For each dataset, we construct dynamic review networks and evaluate each restaurant's survival status in the subsequent year. Here, we present the results for the OpenRice dataset, while those for Yelp are included in Appendix B. To consider the impact of network dynamics, we analyze embeddedness metrics under two scenarios: network embeddedness metrics in each year (Static) and differences in metrics across years (Dynamic).

Distributional Differences. We first use box and line plots to represent the distributional differences in restaurant survival based on the network embeddedness and its temporal evolution on the OpenRice dataset from 2019 to 2023 in Figure 1. The results indicate that both the structural and positional embeddedness of restaurants that tend to close are consistently and significantly lower than those of surviving restaurants, in both static and dynamic terms.

KM Survival Curve. As shown in Figure 2, we then divide all restaurants into high- and low-value groups for each of the two network embeddedness measures and plot their KM survival curves

over the subsequent two years. The results demonstrate that restaurants with lower network embeddedness values or greater declines exhibit significantly lower survival probabilities.

These complementary experiments robustly demonstrate that restaurants' network embeddedness (both structural and positional) and their temporal evolution are positively correlated with survival. This observation further prompts us to explore its significance for restaurant survival prediction models.

3 Method

Motivated by the above analysis, we propose DyE-GNN, a novel dynamic embeddedness-aware GNN model for restaurant survival prediction. As illustrated in Figure 3, DyE-GNN (1) employs distinct aggregation strategies per node type and integrates network embeddedness into restaurant node updates; (2) captures multi-scale temporal dynamics in node evolution; (3) leverages multi-period survival label learning to incorporate extensive restaurant survival knowledge. Here, we first define the problem to be addressed.

3.1 Problem Formulation

Restaurant Survival Prediction. The input consists of a sequence of dynamic review networks with attributes, denoted as $\{\mathcal{G}^T, X_R^T, X_U^T, X_E^T \mid T = T_1, T_2, \dots, T_n\}$, where $X_R \in \mathbb{R}^{|\mathcal{V}_R| \times d_r}$, $X_U \in \mathbb{R}^{|\mathcal{V}_U| \times d_u}$, and $X_E \in \mathbb{R}^{|\mathcal{E}| \times d_e}$ are the attribute matrices for restaurants, users,

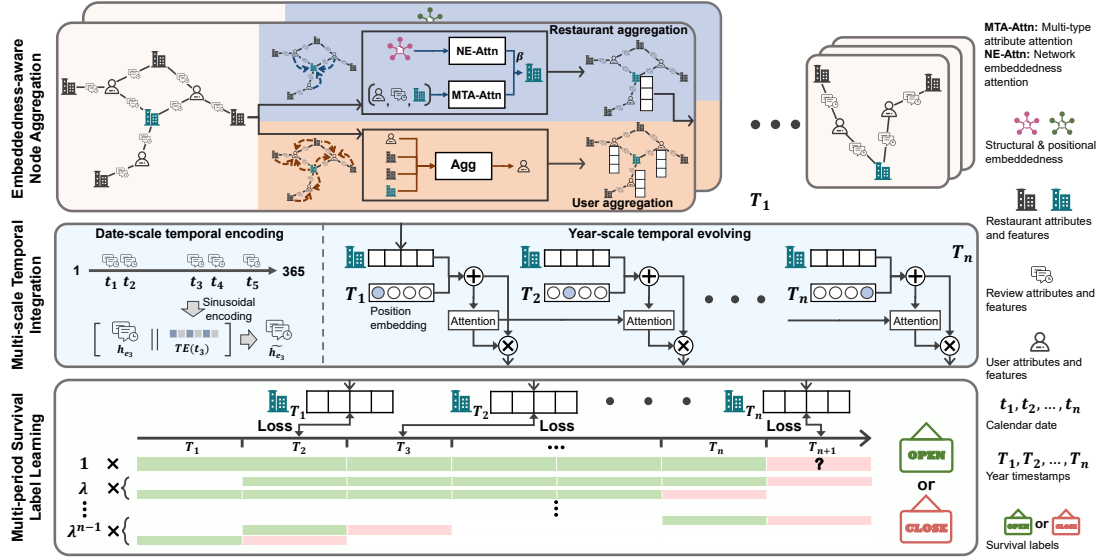


Figure 3: The framework of the dynamic embeddedness-aware GNN model, DyE-GNN, for restaurant survival prediction. It consists of the embeddedness-aware node aggregation module for employing distinct aggregation for each node type and integrating network embeddedness into restaurant nodes, the multi-scale temporal integration module for encoding short-term review fluctuations on a date scale as well as capturing long-term transformations of the review network on a year-scale, and the multi-period survival label learning module for incorporating survival labels across multiple temporal spans to reflect real-world scenarios.

and reviews, respectively. The objective is to predict each restaurant's survival status in the following year, represented by the label set $\mathcal{Y}_R^{\Delta T_{\max}} \in \{0, 1\}$. For training, we use the partial label set $\mathcal{Y}_{\text{train}} \subset \mathcal{Y}_R^{\Delta T_{\max}}$, which is partitioned for the training set, as well as survival labels for other restaurants $\{\mathcal{Y}_R^{\Delta T} \mid \Delta T < \Delta T_{\max}\}$. Then we predict the remaining test labels $\mathcal{Y}_{\text{test}} \subset \mathcal{Y}_R^{\Delta T_{\max}}$.

3.2 Embeddedness-aware Node Aggregation

In this module, we aggregate the attributes of restaurants, users, and reviews to obtain an intra-network representation of each restaurant for every timestamped network. For clarity, time-related markers are omitted in this subsection.

Unlike existing methods that treat restaurant and user nodes identically, we first design distinct aggregation strategies for each node type within the network. From the perspective of restaurant survival, user nodes do not serve as direct samples for the final prediction; instead, their primary contribution arises from their review behaviors, which capture the influence of individual preferences on restaurants. In contrast, restaurant nodes are the primary focus of the prediction task and are influenced by a broader range of factors, including user reviews, competing restaurants, and the overall network structure.

3.2.1 Restaurant Aggregation. For restaurant aggregation, we intend to construct specialized graph attention for multi-type attributes and network embeddedness to represent and integrate diverse information.

- **Multi-type attributes attention** To handle the complex, multi-type attributes of user-restaurant interactions and

their behavior context, we represent each review as a node-edge-node triple. Then we introduce an enhanced graph attention mechanism with edge weights as follows to model dependencies within these triples:

$$\text{cof}_{v_r v_u} = \tilde{a}^T \text{LeakyReLU}(W^R h_{v_r} \parallel W^U h_{v_u} \parallel W^E h_e), \quad (3)$$

$$\alpha_{v_r v_u}^{MTA} = \frac{\exp(\text{cof}_{v_r v_u})}{\sum_{v_{u_i} \in N(v_r)} \exp(\text{cof}_{v_r v_{u_i}})}, \quad (4)$$

where h_e is the review feature between v_r and v_u , and W^E denotes the learnable weight matrix for the feature. $\tilde{a}$ is the shared vector used to compute the attention coefficient. Compared to the traditional graph attention mechanism, our version extracts the linear transformation outside the activation function after concatenating three types of attributes, making the attention coefficients more flexible and responsive to different review behaviors [2].

- **Network embeddedness attention** To emphasize each restaurant node's unique place within the review network, DyE-GNN incorporates a personalized attention mechanism that integrates the two network embeddedness dimensions into the message-passing module:

$$\alpha_{v_r}^{NE} = \theta_{v_r}^l (1 - \theta_{v_r}^l)^l, \quad (5)$$

where $\theta_{v_r}^l$ denotes the normalized embeddedness metric at layer l . We inject structural embeddedness (neighbor out-degree centrality) at the first layer, as it captures a restaurant's local exposure through direct reviewer interactions,

which shallow layers model most effectively. Positional embeddedness (PageRank), reflecting global influence, is integrated from the second layer onward to leverage multi-hop propagation:

$$\theta_{v_r}^l = \begin{cases} \sigma(\log(\text{SE}(v_r) + \epsilon)), & l = 1, \\ \sigma(\log(\text{PE}(v_r) + \epsilon)), & l \geq 2, \end{cases} \quad (6)$$

where ϵ is a small constant for numerical stability.

We then fuse network embeddedness attention with the multi-type attention to update each restaurant node's representation:

$$\alpha_{v_r v_u} = \alpha_{v_r v_u}^{MTA} + \beta \cdot \alpha_{v_r}^{NE}, \quad (7)$$

$$h'_{v_r} = \text{ELU}\left(\sum_{v_{u_i} \in \mathcal{N}(v_r)} \alpha_{v_r v_{u_i}} W_R h_{v_{u_i}}\right), \quad (8)$$

where β is a hyperparameter to modulate the impact of network embeddedness. We introduce a multi-head mechanism for both types of attention to enhance their generalization capability.

3.2.2 User Aggregation. To capture each user node's preferences, we aggregate the feature representations of all restaurants reviewed by that user. At layer $l + 1$, the user embedding is computed as:

$$h_{v_u}^{l+1} = \sigma\left(\sum_{v_{r_i} \in \mathcal{N}(v_u)} W^R h_{v_{r_i}}^l + W^U h_{v_u}^l\right), \quad (9)$$

where $h_{v_r}^l, h_{v_u}^l$ denote restaurant and user features at layer l , W^R, W^U represent the learnable weight matrices for two types of nodes to align different attribute dimensions, $\sigma(\cdot)$ is the ReLU function.

In this module, the initial features for layer 0 comprise restaurant attributes X_R , user attributes X_U , and review attributes X_E . After L layers of message passing, we obtain the final intra-network representations of restaurant nodes $h_{v_r}^{l_n}$.

3.3 Multi-scale Temporal Integration

After obtaining intra-network restaurant representations, we model review dynamics by partitioning temporal information into short-term fluctuations and long-term transformations. This module reflects our preliminary finding that the temporal evolution of network embeddedness correlates strongly with survival. Our ideas can be divided into two opposing yet complementary steps: (1) Encoding date-scale signals to capture event-driven and seasonal variations; (2) Evolving year-scale features to model gradual cyclical transformations.

3.3.1 Date-scale Temporal Encoding. To capture the influence of date-scale review dynamics, we encode each review's calendar date into a continuous scalar and apply sinusoidal encoding [36]. Specifically, in each timestamped network, we normalize the review date to $t_e \in [0, 1]$ by dividing the day-of-year value by 365. We then compute sinusoidal positional encodings as:

$$\begin{aligned} \omega_k &= \exp\left(-\frac{\ln(10000)}{d}k\right), \\ TE(t_e)_{2k} &= \sin\left(t_e \cdot \omega_k\right), k = 0, 1, \dots, \frac{d}{2} - 1, \\ TE(t_e)_{2k+1} &= \cos\left(t_e \cdot \omega_k\right), \end{aligned} \quad (10)$$

This step effectively encodes the temporal sequentiality of dates into similarity to benefit DyE-GNN's feature alignment. Finally, the vector is concatenated with the corresponding review feature as $\tilde{h}_e = h_e \parallel TE(t_e)$ and fused into Equation 3 to allow the restaurant features to integrate short-term variations in reviews.

3.3.2 Year-scale Temporal Evolving. To model long-term and periodic review patterns, we employ a Transformer-based mechanism over annual snapshots. Given the sequence $\{h_{v_r}^{In, T_1}, \dots, h_{v_r}^{In, T_n}\}$, we add position embeddings $\{p^{T_1}, \dots, p^{T_n}\}$ [6] and stack them into $\tilde{h}_{v_r} = \{h_{v_r}^{In, T_1} + p^{T_1}, h_{v_r}^{In, T_2} + p^{T_2}, \dots, h_{v_r}^{In, T_n} + p^{T_n}\}$,

where n is the number of timestamps and d the embedding dimension. We then compute:

$$\tilde{h}'_{v_r} = \sigma\left(\frac{\tilde{h}_{v_r} W^Q (\tilde{h}_{v_r} W^K)^\top}{\sqrt{d}} + \mathbf{M}\right) \tilde{h}_{v_r} W^V. \quad (11)$$

where $W^Q \in \mathbb{R}^{d \times d}, W^K \in \mathbb{R}^{d \times d}, W^V \in \mathbb{R}^{d \times d}$ are used as linear transformations. Significantly, $\mathbf{M}$ is a mask matrix to encode the temporal order, whose elements satisfy $\mathbf{M}_{ij} = 0$ if $i \geq j$, and $\mathbf{M}_{ij} = -\infty$ if $i < j$. This mask enforces a past-to-future information flow by zeroing attention to future timestamps.

3.4 Multi-period Survival Label Learning

Given each restaurant's final representation, DyE-GNN innovatively introduces a multi-period survival label learning for real-world prediction scenarios. Most existing models ignore the contributions of restaurants with varying survival spans, but these heterogeneous samples both shape the review network and provide valuable insights into survival.

Therefore, to capitalize on more survival knowledge, we first split the restaurants into different sets according to the length of their survival during the observation period. For each restaurant $v_r \in \{\mathcal{V}_R^{T_1} \cup \mathcal{V}_R^{T_2} \cup \dots \cup \mathcal{V}_R^{T_n}\}$, let $\Delta T_{v_r} \in [1, n]$ denote its survival duration over the observed period. Then we define $\mathcal{V}_R^{\Delta T}$ to be the set of restaurant nodes partitioned by the value of ΔT , which contains restaurants with the same survival duration. After that, for all restaurant samples under each period length ΔT , we calculate the cross-entropy loss by comparing the survival probability of its model output with the survival label:

$$\ell(p_{v_r}, y_{v_r}) = y_{v_r} \log p_{v_r} + (1 - y_{v_r}) \log(1 - p_{v_r}) \quad (12)$$

$$L_{\Delta T_{\max}} = - \sum_{\substack{v_r \in \mathcal{V}_R^{\Delta T_{\max}} \\ y_{v_r} \in \mathcal{Y}_{\text{train}}}} \ell(p_{v_r}, y_{v_r}), \quad (13)$$

$$L_{\Delta T} = - \sum_{\substack{v_r \in \mathcal{V}_R^{\Delta T} \\ \Delta T < \Delta T_{\max}}} \ell(p_{v_r}, y_{v_r}), \quad (14)$$

where $\Delta T_{\max}$ means surviving throughout the observation period and y_{v_r} indicates the survival label. The p_{v_r} denotes the predicted probability of restaurant closure, obtained from the last timestamp feature after the year-scale evolution.

Finally, we merge these losses to train the entire DyE-GNN. We introduce a decay factor λ to mitigate noise from restaurants with

limited operating durations:

$$L = L_{\Delta T_{\max}} + \sum_{\Delta T < \Delta T_{\max}} \lambda^{\Delta T_{\max} - \Delta T} L_{\Delta T}. \quad (15)$$

3.5 Computational Complexity

To verify the applicability of DyE-GNN, we discuss the time complexity of the model. In the embeddedness-aware node aggregation model, the network embeddedness metrics are computed with complexity $O(|\mathcal{V}_R|) + O(T_p \cdot |\mathcal{E}|)$, where T_p denotes the number of PageRank iterations. And the complexity for user aggregation and restaurant aggregation can be expressed as $O(d(|\mathcal{V}_U| + |\mathcal{V}_R|) + (d + d_t)|\mathcal{E}|) + O(d(|\mathcal{V}_U| + |\mathcal{E}|/2)) + O(3d|\mathcal{E}|)$, where d_t is the length of the calendar date encoding. Considering timestamps and multiple neural network layers L , the complexity of the whole module can be organized as $\sum_{i=1}^n O(2|\mathcal{V}_U^{T_i}| \cdot L \cdot d + |\mathcal{V}_R^{T_i}|(1 + L \cdot d) + (T_p + L \cdot (\frac{3}{2}d + d_t))|\mathcal{E}^{T_i}|)$. For the Multi-scale Temporal Evolution module, given $N_R = |\mathcal{V}_R^{T_1} \cup \mathcal{V}_R^{T_2} \cup \dots \cup \mathcal{V}_R^{T_n}|$ as the number of restaurants in all timestamps, the time complexity is described as $O(N_R \cdot n^2 \cdot d)$. After combining the complexity $O(N_R)$ of the last module, the complexity of the whole model can be integrated to $O(\sum_{i=1}^n [2|\mathcal{V}_U^{T_i}| \cdot L \cdot d + |\mathcal{V}_R^{T_i}|(1 + L \cdot d) + (T_p + L \cdot (\frac{3}{2}d + d_t))|\mathcal{E}^{T_i}|] + N_R \cdot (n^2 \cdot d + 1))$. From the analysis, User nodes that are much larger than the number of restaurants only participate in the calculation during the node aggregation, while the subsequent calculation only considers restaurant samples. This can effectively take into account the user's impact on the reviews and save the computational complexity of the subsequent steps.

Then we describe the whole procedure of the model algorithm in the Appendix D.

4 Evaluation

4.1 Experimental Setup

4.1.1 Datasets. Prior studies have been limited by inaccurate closure labels, which are often inferred from the timestamp of a restaurant's last review [18, 43]. To address this limitation, we compile a rigorously annotated OpenRice dataset that includes official closure dates from the Food and Environmental Hygiene Department of Hong Kong. This dataset comprises 5,483 restaurants from Hong Kong's Yau Tsim Mong district, with 444,531 reviews spanning from 2019 to 2023. Additionally, we construct a Yelp dataset containing 1,909 restaurants and 226,840 reviews from Boston spanning 2014 to 2018. Table 1 lists statistics for all relevant data, including preliminary and model evaluation results. For both datasets, we have compiled restaurant, review, and user attributes, as well as survival labels (Details in Appendix C).

4.1.2 Baselines. We choose the baseline methods in four categories, which are sequential models, static GNN, continuous-Time Dynamic Graph (CTDG) models, and discrete-time dynamic graph (DTDG) models.

Sequential models. First, we directly consider a multi-year restaurant attributes as a sequence and input them into two classical sequential neural network models: RNN [4] processes sequential

data by introducing hidden states that propagate across time steps. They can naturally handle variable-length sequences and retain contextual memory. LSTM [11] addresses the vanishing-gradient problem in RNNs over long sequences by incorporating gating mechanisms, enabling them to effectively capture long-range temporal dependencies.

Static GNN. Second, we construct all review data for the entire period as a single network and disregard the temporal information of each review to deploy the baseline methods of the static GNN category. This contains four classical GNNs methods and two recent recommendation systems methods: GCN [16] is the typical GNN method that aggregates features from its direct neighbors and itself. GraphSage [9] is an inductive GNN method that generates node embeddings by sampling and aggregating features from a fixed-size neighborhood of each node. GAT [37] uses the attention coefficient to aggregate node features. GATv2 [2] improves the original attention computation of GAT and further enhances the flexibility and expressiveness of the attentional mechanism. LightGCN [10] is a lightweight graph convolution network designed for recommender system tasks, which omits feature transformations and nonlinear activation operations. ApeGNN [42] proposes an adaptive information transfer mechanism on LightGCN to enhance the modeling of complex graph structures. We choose the personalized PageRank (PPR) idea as the weighting coefficient calculation to contrast our network embeddedness idea.

CTDG models. Third, based on the static GNN setup, we additionally consider date-scale temporal information in a single network to utilize several continuous-time dynamic GNN methods. In these methods, we combine and normalize calendar date information for multiple year periods to distinguish between dates for different years. We choose the two most relevant to our model within this class of methods: TGAT [39] embeds temporal information directly into the graph attention mechanism, which in turn captures the interaction dynamics between nodes evolving over time. BiDyn [38] utilizes two time information channels, forward and backward, respectively, to capture the dynamics of nodes. DyGFormer [41] further advances continuous-time dynamic graph learning through a Transformer-based architecture that learns solely from nodes' historical first-hop interactions.

DTDG models. Fourth, we deploy the discrete-time dynamic GNN methods in dynamic restaurant review networks. These methods tend to disregard the calendar dates of the reviews in each timestamped network and only consider the year-scale temporal information. We deploy two representative DTDG methods and modify a state-of-the-art static GNN model to accommodate discrete-time dynamic graphs: EvolveGCN [26] utilizes recurrent neural networks to update the graph convolutional layer weights to automatically adjust the model's parameters in dynamic graphs. DySAT [32] uses a self-attentive mechanism to capture both network structure information and temporal information. DTFormer [3] replaces the traditional GNN+RNN pipeline with a unified attention-based encoder. ApeGNN_{dyn} is a model that improves ApeGNN for dynamic restaurant review networks. We add the Year-scale time evolving module to it to pass information between dynamic networks.

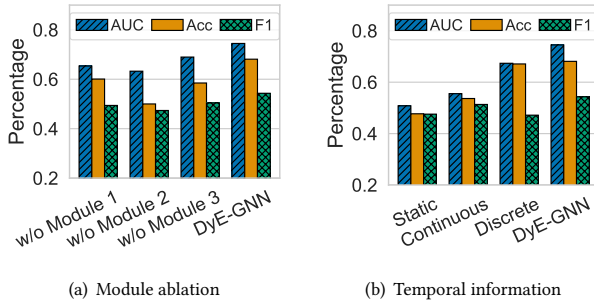
4.1.3 Metrics. In order to sufficiently validate our model, we select three persuasive metrics: AUC, Acc, F1. Among them, Acc can

Table 1: Yearly statistics for OpenRice and Yelp datasets

Category		OpenRice							Yelp							
		2019	2020	2021	2022	2023	2024	Total	2014	2015	2016	2017	2018	2019	2020	Total
Restaurant	Opened	522	424	522	455	373	–	–	171	176	178	165	177	–	–	–
	Closed	–	344	352	307	335	486	–	–	135	126	116	146	147	258	–
User	Active	3,708	3,788	3,958	4,106	4,144	3,658	5,483	1,299	1,340	1,392	1,441	1,472	1,325	1,067	1,909
	Review	20,772	18,220	22,803	21,878	23,427	–	63,826	21,675	24,543	27,078	29,073	30,846	–	–	112,973
		86,206	65,211	93,029	95,883	104,202	–	444,531	39,227	42,614	46,095	48,091	50,813	–	–	226,840

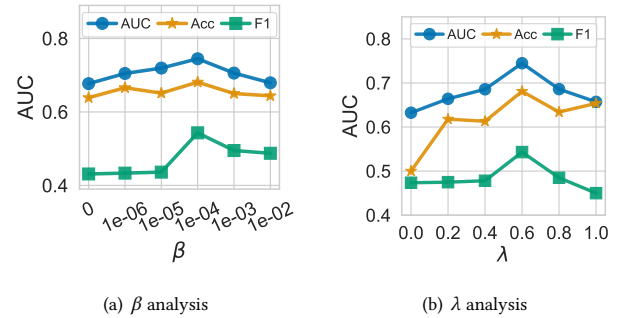
Table 2: Performance comparison in predicting restaurant survival.(%)

Category	Method	OpenRice						Yelp					
		50%			80%			50%			80%		
		AUC	Acc	F1	AUC	Acc	F1	AUC	Acc	F1	AUC	Acc	F1
Sequential models	RNN [4]	54.74	50.12	50.09	61.82	51.91	51.31	54.31	54.63	49.90	59.41	54.19	51.14
	LSTM [11]	57.87	49.76	49.74	62.60	50.89	49.99	57.25	50.20	19.82	65.28	50.65	21.79
Static GNNs	GCN [16]	54.13	52.38	40.53	63.33	59.94	35.92	56.80	54.42	34.94	67.29	65.84	32.49
	GraphSage [9]	59.73	56.04	43.53	59.96	61.67	47.19	53.72	50.00	6.10	61.28	50.00	6.09
	GAT [37]	54.24	50.00	5.71	60.47	50.00	5.65	55.70	50.00	6.10	60.93	55.20	16.12
	GATv2 [2]	56.85	50.00	5.71	60.38	50.00	5.71	57.02	50.00	6.10	59.94	52.97	12.03
	LightGCN [10]	61.53	57.68	21.46	59.96	52.93	44.00	53.91	50.00	6.10	56.12	53.96	42.49
	ApeGNN [42]	62.82	56.21	19.07	62.19	61.12	37.32	58.70	55.97	30.83	62.06	53.01	46.56
CTDG models	TGAT [39]	57.68	54.43	17.69	62.10	55.59	20.37	50.46	51.49	50.72	57.71	53.64	49.41
	BiDyn [38]	54.85	52.64	45.34	60.07	49.33	48.12	51.87	50.63	49.26	57.14	50.50	7.12
	DyGFormer [41]	60.82	56.75	47.03	60.84	50.00	47.36	52.15	52.16	50.19	63.15	62.27	54.32
DTDG models	EvolveGCN [26]	63.03	54.71	34.81	70.78	65.25	46.80	56.31	48.23	45.68	63.70	50.00	6.09
	DySAT [32]	63.26	57.62	45.98	71.16	62.73	50.50	58.85	54.17	49.98	68.25	55.09	21.41
	DTFormer [3]	61.03	58.44	48.30	67.77	63.79	52.58	53.78	57.46	45.64	61.99	54.03	51.92
	ApeGNN _{dyn}	64.79	60.84	47.83	69.32	61.38	48.98	60.20	57.42	49.42	72.00	63.55	54.75
Ours	DyE-GNN	70.88	63.35	51.00	74.48	68.11	54.32	67.23	59.19	51.80	77.19	63.51	56.28

**Figure 4: Ablation analysis of DyE-GNN on the OpenRice dataset.**

directly assess the model’s prediction ability for samples. Since the number of surviving and closed restaurants is not balanced, we choose Macro F1 to directly obtain the mean F1 score for both positive and negative samples to eliminate the effect of sample imbalance. We repeat each experiment five times and report all metrics for the highest test AUC value.

4.1.4 Implementation Details. All experiments were run on a machine equipped with the NVIDIA H100 GPUs and around 80GB of VRAM. Except for the parameters specific to the original paper, the epoch is set to 200, the hidden dimension is set to 256, the multi-heads number is set to 4, the number of layer is set to 2, and the dropout is set to 0.1 on the OpenRice dataset and 0.02 on the Yelp dataset. In addition to this, we empirically set the learning rate to 0.0001 and divide the training set ratios into two cases, 80% and

**Figure 5: Parameter analysis of DyE-GNN on the OpenRice dataset.**

50%, to report performance comparisons. To mitigate the issue of imbalanced positive and negative samples, we performed negative sampling [40] for surviving restaurants, to remove some of the samples whose node attributes are most similar to positive samples. It is set to 0.4 in the OpenRice dataset, and 0.2 in the Yelp dataset. We repeat each experiment five times and report all metrics for the time with the highest test AUC value.

4.2 Performance Comparison

Table 2 reports the performance comparison of various methods for restaurant closure prediction on both the OpenRice and Yelp datasets. Several key conclusions can be drawn: (1) ApeGNN consistently outperforms LightGCN, indicating that integrating person-alized centrality substantially enhances restaurant survival

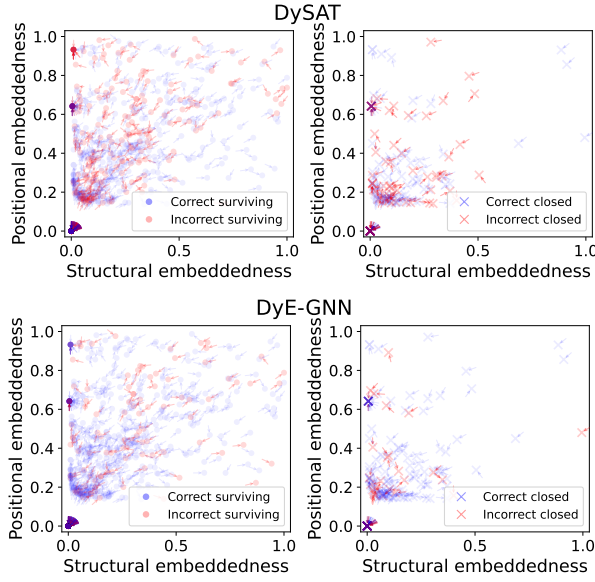


Figure 6: Visualization of restaurant nodes' network embeddedness in correlation with prediction results. The horizontal and vertical axes represent normalized embeddedness values after trimming outliers. Each line attached to a point indicates the change from 2019 to 2023: its direction shows whether the metrics increased or decreased, and its length reflects the magnitude of change. Point shapes denote next-year survival (circle = survived; cross = closed), and point colors indicate prediction correctness (blue = correct; red = incorrect).

prediction; (2) DTDG baselines surpass all other baseline categories, highlighting the critical role of modeling network dynamics for this task; (3) Overall, DyE-GNN achieves the highest performance across different training ratios and prediction scenarios, illustrating the superiority of our approach.

Due to space limitations, all experiments below are shown using the OpenRice dataset, while experiments using the Yelp dataset can be found in Appendix E.

4.3 Ablation Analysis

Module Ablation. To assess each module's impact, we evaluate three DyE-GNN variants (Figure 4(a)): (1) w/o Module 1: Replacing module 1 to the vanilla GCN; (2) w/o Module 2: making predictions in a single review network; (3) w/o Module 3: no labels learning of other periods. As shown in the figure, omitting Module 2 yields the greatest performance degradation, followed by Module 1; nevertheless, all modules contribute significantly.

Temporal Information. To explore the contribution of date-scale encoding and year-scale evolving to DyE-GNN, Figure 4(b) compares three time-aware DyE-GNN variants: static, date-scale only (Continuous), and year-scale only (Discrete). The results confirm the effectiveness of multi-scale temporal integration in enhancing restaurant survival prediction performance, with year-scale evolution proving more important than date-scale embedding.

4.4 Parameter Analysis

Network Embeddedness Weight. To effectively incorporate network embeddedness into DyE-GNN, we introduce β that adjusts its contribution relative to multi-type attribute attention in node aggregation. As shown in Figure 5(a), completely removing network embeddedness attention ($\beta = 0$) significantly degrades model performance, while setting β to the order of magnitude of 1×10^{-4} yields optimal results.

Decay Factor of Other Periods. To capture survival knowledge from restaurants with varying lifespans while minimizing unnecessary noise, we use a decay factor λ to modulate the label learning for these restaurants. As illustrated in Figure 5(b), both ignoring and overemphasizing this survival knowledge result in poorer model performance, whereas setting λ to 0.6 achieves the optimal balance between knowledge utilization and prediction accuracy.

4.5 Embeddedness vs. Prediction Visualization

Figure 6 illustrates the relationship between network embeddedness (and its temporal evolution) and prediction outcomes, using scatter plots of structural and positional embeddedness for all restaurants in the 2023 review network. Two main observations emerge from this analysis. First, when comparing the figures horizontally, closed restaurants (crosses) tend to cluster near the origin, with their trend lines generally pointing inward, corroborating our preliminary findings. Second, DyE-GNN outperforms DySAT in identifying closures, for restaurants with low and decreasing embeddedness, demonstrating its superior ability to identify at-risk restaurants at the network margin.

5 Conclusion

This study pioneeringly bridges sociological network embeddedness theory and dynamic graph neural network design for restaurant survival prediction. After demonstrating that review network embeddedness positively correlates with survival outcomes, we introduce DyE-GNN, a dynamic embeddedness-aware GNN that introduces network embeddedness theory to guide the model design and leverages domain knowledge to enable robust adaptability. We validate DyE-GNN's superior performance on both the newly curated OpenRice dataset and the widely used Yelp.

Our findings offer practical insights for restaurant practitioners. First, enhancing a restaurant's network embeddedness can reduce its risk of closure. Strategies include increasing restaurant exposure, engaging highly active and influential reviewers, and maintaining positive trends in review dynamics. Second, by continuously monitoring embeddedness metrics and temporal patterns, DyE-GNN can function as an early-warning system, enabling timely adjustments in marketing, staffing, and menu strategies to reduce closure risk.

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A Related Work

A.1 Business Survival Prediction

To position our model within the broader literature on business survival prediction, we conduct a comparative analysis of the influencing factors and methods explored in these studies.

A.1.1 Influencing Factors. The factors influencing survival discussed in the existing literature can be categorized into four categories: firm characteristics, competition, macroconditions, and user-generated content [43]. Business survival depends on firm characteristics, such as firm age [5], ownership structure [13, 21, 28, 29], business model [29], and operational practices [1, 14, 21, 27, 30, 35].

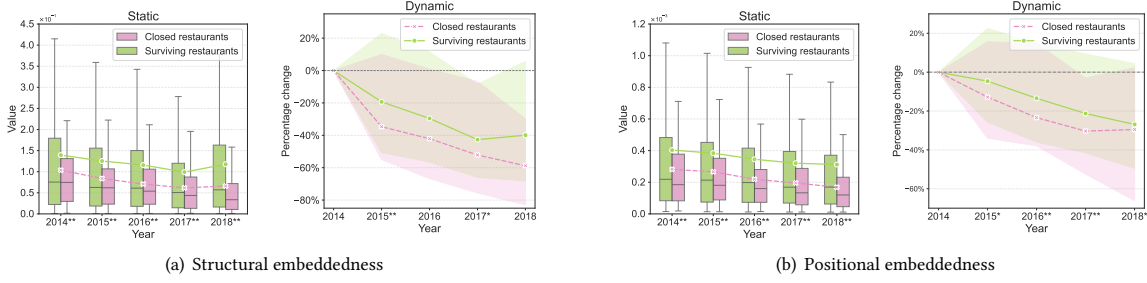


Figure 7: The distribution of network embeddedness for restaurants with different survival scenarios on the Yelp dataset.

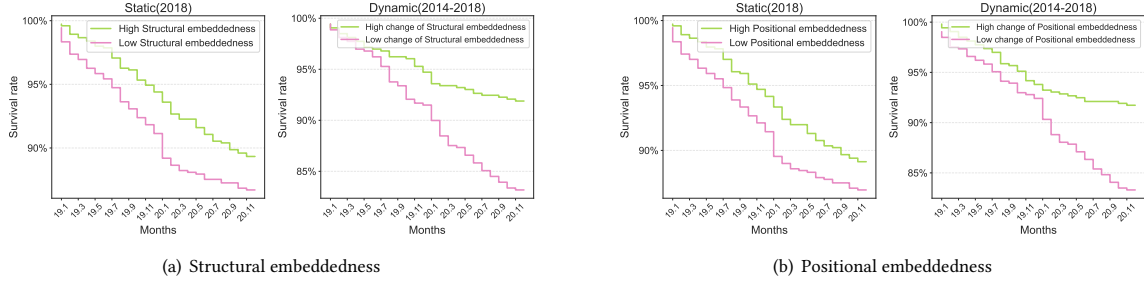


Figure 8: The effect of network embeddedness on restaurants' KM survival curves on the Yelp dataset.

Moreover, it is shaped by peer-competitor concentration [5, 29] and macro-level market conditions [5, 8, 12, 27, 28, 30]. As social media proliferates, online reviews have become critical to understanding restaurant performance. Review data, from ratings and votes to text comments and images, encapsulates rich insights into survival prospects. User ratings and votes provide direct evaluations that serve as key predictors in survival analyses [18, 24, 34, 43]. In addition, by applying natural language processing tools, the text content of reviews can also express users' sentiment towards restaurants [17–19, 22, 34]. Furthermore, review photos, as information that can be easily processed by humans, contain even richer potential value for survival prediction to be processed by computer vision methods [43]. However, beyond these content-level indicators, the structural patterns of restaurant–user interactions remain largely unexplored. It is obvious that every restaurant is embedded in a vast review network, but the mechanisms by which network position influences survival have received little attention. To address this gap, we leverage network embeddedness theory and statistical analysis to demonstrate that both static and dynamic network embeddedness metrics correlate with restaurant survival.

A.1.2 Methods. Early studies on business survival prediction primarily employed econometric models (e.g., logit and hazard models) to measure bankruptcy risk based on firm characteristics, competitive conditions, and macroeconomic factors [1, 5, 29]. More recent studies have incorporated machine learning classifiers, such as SVM, XGBoost, Gradient-Boosted Decision Trees (GBDT), and Hidden Markov Model (HMM), to leverage review-level attributes like ratings and votes to achieve better prediction results [20, 24]. Deep learning models (such as language models, CNNs, and LSTMs) have further advanced text- and image-based survival prediction [34, 43].

Despite these advances, existing methods all neglect the network embeddedness factors discussed above. Although Li et al. applied co-attention and static GNN to model restaurant–user interactions, they ignored the temporal evolution of the review network [19]. Dynamic network structures allow models to more accurately reflect real-world review patterns. Therefore, we propose DyE-GNN, a dynamic GNN model that jointly captures the semantic richness of survival-related attributes and the evolving topology of the review network in an end-to-end framework. In the next subsection, we review the development of dynamic GNN models and situate DyE-GNN within this landscape.

B Preliminaries on the Yelp Dataset

As shown in Figures 7, 8, we repeated all preliminary experiments on the Yelp dataset from 2014 to 2018 and found consistent results.

C Dataset Details

C.1 Restaurant Attributes

For OpenRice, 33 features across four categories (macro, restaurant, competition, and review) were extracted (Table 3), including spatial coordinates, spending, cuisine type, and yearly dynamics (e.g., restaurant age, competitor count, prior reviews). For Yelp, 19 dimensions were extracted (Table 4) following the same logic, covering category, price, rating, and review statistics.

C.2 Review Attributes

OpenRice reviews include 23 dimensions in five groups—rating, votes, text, photo, and time (Table 5). Yelp reviews contain 15 attributes (Table 6), with additional BERT-based aspect sentiment

Table 3: Illustration of restaurant attributes on OpenRice dataset

Category	Attribute	Dimension	Is dynamic	Specification
Macro	Subdistrict code	1	✗	Subdistrict codes under the Yau Tsim Mong of Hong Kong
	Latitude	1	✗	
Restaurant	Longitude	1	✗	
	Spending	1	✗	
	Votes	3	✗	Votes for good, ok, and bad
	Cuisine type	11	✗	Encoding for Top10 cuisines and others
	Age	1	✓	
Competition	Number of Competitors	1	✓	Number of restaurants sharing the same cuisine in the same zip code
Review	Number of previous reviews	1	✓	Number of reviews before the year
	Number of previous review photos	1	✓	Number of photos before the year

Table 4: Illustration of restaurant attributes on Yelp dataset

Category	Attribute	Dimension	Is dynamic	Specification
Macro	Zip code	1	✗	Zip code of the area where the restaurant is located
Restaurant	Latitude	1	✗	
	Longitude	1	✗	
	Is chain	1	✗	
	Price	1	✗	
	Cuisine type	11	✗	Encoding for Top10 cuisines and others
Competition	Age	1	✓	
	Number of Competitors	1	✓	Number of restaurants sharing the same cuisine in the same zip code
Review	Number of previous reviews	1	✓	Number of reviews before the year
	Number of previous review photos	1	✓	Number of photos before the year

Table 5: Illustration of review attributes on OpenRice dataset

Category	Attribute	Dimension	Specification
Rating	Taste	1	
	Decor	1	
	Service	1	
	Hygiene	1	
	Value	1	
Vote	Dining experience	3	Good, ok, or bad
Content	Text length	2	Length of title and content
	Number of Emojis	2	Number of emojis in the title and content
	Number of Emotional Emojis	6	Number of emojis expressing positive, neutral, or negative emotions in title and content
	Number of replies	1	Number of replies to this review
Photo	Number of photos	1	
Time	Year	1	Temporal information is divided into two parts: year and calendar date
	Calendar date	1	

scores for taste, service, location, price, and atmosphere. We compute user attributes by taking a weighted average of their review attributes within each yearly snapshot.

Table 6: Illustration of review attributes on Yelp dataset

Category	Attribute	Dimension	Specification
Rating	Stars	1	
Votes	Useful votes	1	
	Funny votes	1	
	Cool votes	1	
	Total votes	1	Sum of above 3 votes
Content	Text length	1	
	Tastiness sentiment	1	Calculated by BERT-based ABSA, same as below
	Service sentiment	1	
	Location sentiment	1	
	Price sentiment	1	
	Atmosphere sentiment	1	
Photo	Number of photos	1	
Time	Year	1	Temporal information is divided into two parts: year and calendar date
	Calendar date	1	

Algorithm 1 Algorithm of DyE-GNN

Require: Dynamic review networks $\mathcal{G}^{dyn}$; **Parameters:** β, λ
Ensure: Restaurant survival probability p_{v_r}

- 1: **for** each timestamp T_i **do**
- 2: Network embeddedness calculation by Eqs. (1), (2).
- 3: **for** each layer l in L **do**
- 4: **for** each restaurant node v_r in $\mathcal{V}_R^{T_i}$ **do**
- 5: Date-scale time encoding by Eq. (10).
- 6: Multi-type attributes attention by Eq. (4).
- 7: Network embeddedness attention by Eq. (5).
- 8: Restaurant aggregation by Eqs. (7), (8).
- 9: **end for**
- 10: **for** each user node v_u in $\mathcal{V}_U^{T_i}$ **do**
- 11: User aggregation by Eq. (9).
- 12: **end for**
- 13: **end for**
- 14: **end for**
- 15: **for** each restaurant node v_r in $\mathcal{V}_R^{T_1} \cup \mathcal{V}_R^{T_2} \cup \dots \cup \mathcal{V}_R^{T_n}$ **do**
- 16: Year-scale time evolution by Eq. (11).
- 17: **end for**
- 18: Cross-entropy loss calculation by Eqs. (13), (14).
- 19: Train parameters through the loss in Eq. (15).
- 20: **Output:** Restaurant survival probability p_{v_r} .

C.3 Survival Labels

For OpenRice, we obtained reliable survival labels by linking each restaurant to its official license data from the Food and Environmental Hygiene Department of Hong Kong. The start and end dates of business licenses were matched (via fuzzy name and address matching) to determine the precise closure time of each restaurant, ensuring objectivity and eliminating bias from review sparsity. For **Yelp**, we identified closure events using review text signals following Zhang et al. [43]. Closure-related keywords were detected to infer closing years, and restaurants without such signals were assumed to remain open until their latest review. This hybrid text-temporal method provides robust yet interpretable closure estimation.

D Algorithm

We describe the whole procedure of the model algorithm in Algorithm 1. For each epoch of the training process, after passing

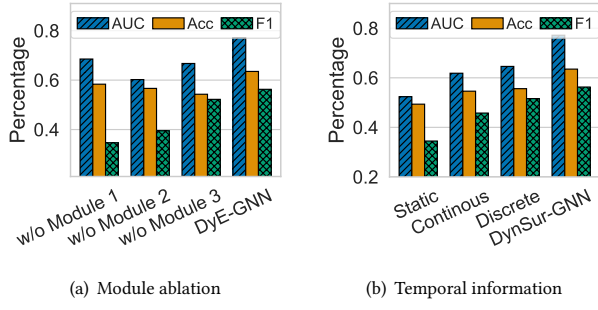


Figure 9: Ablation analysis of DyE-GNN on the Yelp dataset

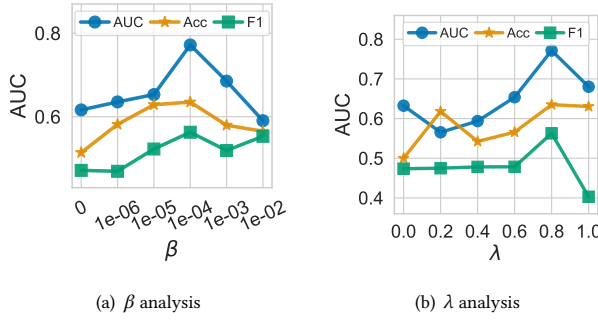


Figure 10: Parameter analysis of DyE-GNN on the Yelp dataset

through the three modules in turn, DyE-GNN updates all model parameters according to the final loss and gets the restaurant survival probability.

E Supplementary Experiments on the Yelp Dataset

As shown in Figures 9, 10, and 11, we report ablation analysis, parameter analysis, and visualization on the Yelp dataset. Most of the experimental conclusions are similar to those of OpenRice. The difference is that on the Yelp dataset, the decay factor for other periods λ is set to 0.8, and the negative sampling ratio is set to 0.2 for best results.

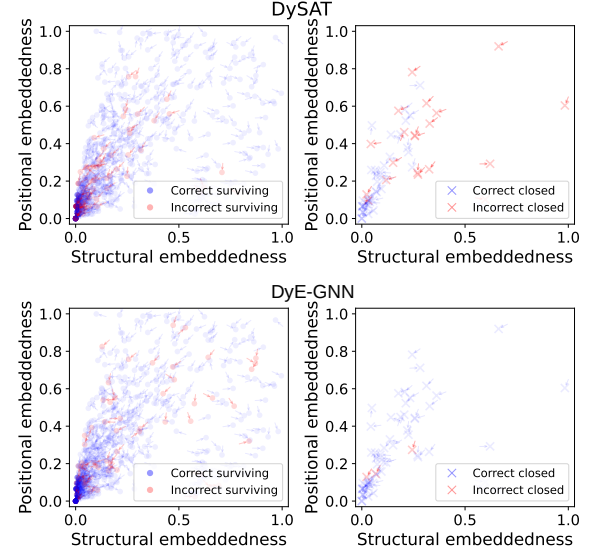


Figure 11: Visualization of restaurant nodes' network embeddedness in correlation with prediction results on the Yelp dataset.